

Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study

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Abstract

Conformal prediction coverage guarantees rely on exchangeability but can degrade severely under shift. We ask which deployed models fail before any test data is observed. On a temporal COVID-19 split of SALT supply-chain tasks and extensions to 11 additional domains, we find that *SHAP concentration*—the fraction of SHAP mass on the top feature—strongly predicts coverage failure for gradient-boosted classifiers under covariate shift. Concentration alone explains why coverage losses range from near-zero to over 70% despite similar shift. We report 50-seed experiments, confirm a concentration–degradation relationship across 16 multiclass tasks (Spearman $\rho = 0.853$, $p < 0.001$), and show that standard shift detectors (MMD/C2ST/PSI) do not predict failure severity. A short exploratory decision rule uses SHAP concentration with a 40% threshold and conservative ambiguity band. This work is associational: it identifies a model-specific pre-deployment risk signal tied to failure mechanism, not a universal causality law.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Computer systems organization** → **Cloud computing**.

Keywords

Conformal prediction, covariate shift, SHAP, robustness, uncertainty quantification

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1 Introduction

Conformal prediction offers finite-sample coverage guarantees under exchangeability, not under arbitrary shift. In practice, shift is common, while practitioners need a *pre-deployment* screen that flags likely breakdown before deployment. Conformal extensions under covariate and adaptive settings provide useful guarantees under assumptions [1, 9, 10], but do not by themselves provide a task-level pre-deployment severity signal. We evaluate whether the concentration of feature reliance in a model is such a screen.

2 Related Work

Shift detectors can identify distribution drift but may not prioritize which models fail severely [3, 4, 7]. APS and adaptive conformal methods capture performance under non-stationarity [2, 8], while attribution methods provide model-level interpretability [5, 6]. We compare these ideas by testing whether attribution structure predicts *severity* of conformal degradation before test-time data is observed.

Contributions.

- We evaluate a model-level diagnostic, *SHAP concentration*, as an early-warning metric before test observation.
- We prove a monotone score-inflation mechanism for APS under high concentration on a shifted feature.
- We validate across 8 SALT tasks with heavy temporal shift, then extend to 16 multiclass tasks across 9 domains.
- We benchmark against baseline diagnostics and show that shift detectors predict shift existence but not degradation severity.

3 Setup

3.1 Data and Splits

We use the SALT benchmark (8 classification tasks) with aligned temporal splits:

- Train: before Feb 2020, Val: Feb–Jul 2020, Test: after Jul 2020.

For external tests, we use 11 datasets across 10 domains.

We compute SHAP concentration only from validation data:

$$C = \frac{\phi_1}{\sum_{j=1}^p \phi_j}, \quad (1)$$

where ϕ_j is mean absolute SHAP for feature j and ϕ_1 is top-ranked feature SHAP. All models use the same pipeline with 50 random seeds; we report mean coverage, 95% CI, and Wilcoxon tests.

3.2 Conformal Protocol

We use Adaptive Prediction Sets (APS) with $\alpha = 0.1$ and target coverage $1 - \alpha = 0.9$. We use a non-randomized variant. We run 50 seeds for all SALT tasks and 10–50 seeds where available for external validation.

4 Why Concentration Could Matter

Two tasks share near-zero feature overlap, yet one loses little coverage and another loses over 70%. This indicates that shift size alone is insufficient and that dependence geometry matters. Jaccard overlap is a coarse proxy, while concentration distinguishes how concentrated the model’s dependence becomes on dominant features. Concretely, if a shifted feature

absorbs mass in the prediction ranking, APS can become more conservative on the true label despite similar observed turnover.

5 Theory

For a K -class classifier, APS score is the mass assigned at or above the true label.

THEOREM 1 (SCORE INFLATION UNDER CONCENTRATED SHIFT). *Consider*

$$\hat{p}(y | x) = C \cdot g(y | x_1) + (1 - C) \cdot h(y | x_{\setminus 1}), \quad K \geq 3,$$

with concentrated misclassification on x_1 under shift: $g(y^* | x_1^{test}) \leq \epsilon < 1/K$. If the residual term is exchangeable between calibration and test, then APS scores shift to larger values as C increases and coverage upper bounds are non-increasing in C .

Theorem 1 formalizes a mechanism: more mass on one shifted feature increases true-label score requirements, so prediction sets grow and coverage degrades. A full derivation appears in the paper code repository and experimental notebook.

6 Empirical Findings

6.1 SALT Tasks: large variation under the same shift

Despite identical COVID-era shift, coverage drops span 0.1–77.1%, and concentration tracks these failures: catastrophic outcomes are concentrated, robust tasks are not.

6.2 Concentration diagnostics

SHAP concentration is strongest among evaluated pre-deployment diagnostics for these experiments.

6.3 Mechanism checks

Three observations support mechanism-level interpretation:

- (1) Catastrophic tasks show strong right-shifts in APS score CDFs (higher scores needed for the true label).
- (2) SHAP concentration is stable under resampling (within-task coefficient variation below 1%).
- (3) Seed instability appears mostly in edge cases where top feature identity shifts, not in robust low-concentration tasks.

7 Decision Guidance

We propose an exploratory protocol:

- **Compute** SHAP concentration on validation data.
- If $C \leq 40\%$, treat as lower-risk.
- If $C > 40\%$, check for protective secondary features and model-specific stability.
- In ambiguity band 35–45% or unstable concentration, use conservative monitoring and do not assume robustness.

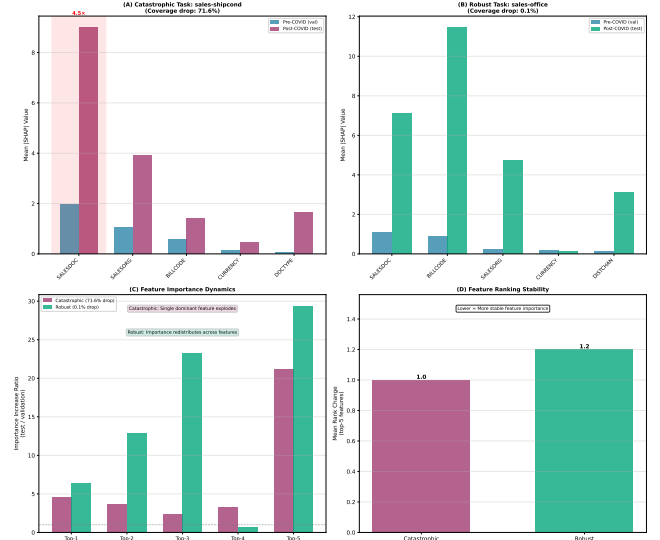


Figure 1: Representative SHAP concentration dynamics under shift. Concentrated single-feature dependence leads to catastrophic coverage loss, while dispersed importance is safer.

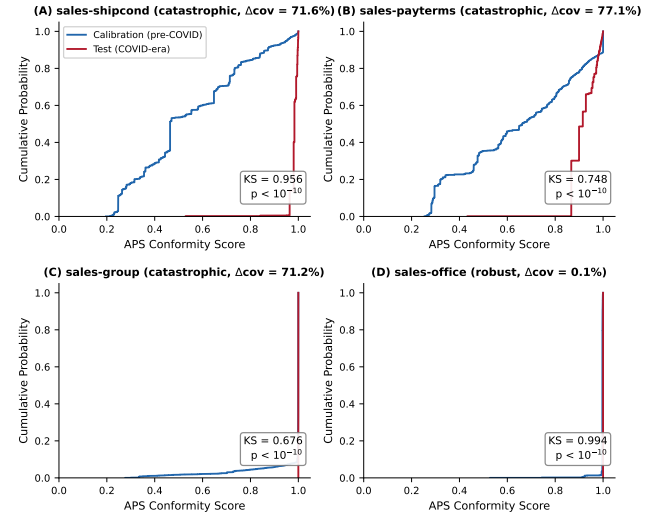


Figure 2: Conformity score CDFs under calibration and test (seed=42). Right-shifts in catastrophic settings mean higher score requirements for the true label.

For intermediate and high C , retraining can recover part of the loss in some tasks, and ACI/RAPS can be complementary but not universally effective.

8 External Validation

In 9 external held-out domains (16 multiclass tasks total), concentration remains associated with failure severity. Covertypes

Table 1: Coverage degradation on SALT (50 seeds).

Task	#Cls	Val Coverage	Test Coverage	Drop	SHAP C
s-shipcond	45	93.5	21.8	71.6	50.7
s-group	462	83.6	12.4	71.2	47.3
s-payterms	135	90.8	13.7	77.1	54.2
i-plant	35	92.0	81.4	10.6	23.9
i-shippoint	70	91.2	72.7	18.5	48.8
s-incoterms	13	95.5	87.0	8.5	23.7
i-incoterms	13	95.0	83.7	11.3	28.9
s-office	25	99.9	99.9	0.1	42.6

Table 2: Diagnostics across SALT and external multiclass tasks.

Set	n	Corr. with drop (Spearman ρ)	p	Significance
SALT multiclass ($K \geq 3$)	8	0.833	0.010	<i>sig.</i>
SALT + cross-domain multiclass	16	0.853	< 0.001	<i>significant</i>
Native FI concentration	8	0.667	0.071	<i>weak</i>
Ensemble disagreement	8	0.452	0.260	<i>weak</i>
MMD / C2ST / PSI	8	≤ 0.19	> 0.6	<i>not predictive</i>

is consistently catastrophic with high concentration; many low-concentration datasets remain robust. Boundary cases (e.g., KDDCup99) show intermediate variability, motivating uncertainty bands and post-hoc calibration.

9 Discussion and Limitations

Coverage degradation risk is model-specific and mechanism-dependent. SHAP concentration explains when failure is amplified through concentrated dependence, but does not replace full monitoring. Evidence is strongest for gradient-boosted trees and current seeds; transfer across model classes is partial. **Limitations.** Current evidence is strongest for multiclass tasks and this experimental protocol; binary settings and some model classes may behave differently. Exploratory thresholds should be revalidated before deployment and are not intended as a universal rule.

10 Conclusion

Across extensive 50-seed experiments and external validation, SHAP concentration is a practical pre-deployment indicator of conformal vulnerability under shift. We provide an exploratory diagnostic and a conservative thresholding workflow; we deliberately avoid causal claims and avoid over-generalization beyond the observed mechanism.

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